

# Adaptive Verifier Allocation for Efficient Test-Time Compute Scaling: A Regret-Theoretic Analysis

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## Abstract

We study the problem of efficient verifier allocation during test-time compute scaling for large language models. Recent work has established that process reward models (PRMs) used as verifiers are asymptotically optimal for scaling test-time compute; however, invoking a verifier at every node of a search tree incurs costs that are linear in the search budget, limiting scalability. We introduce *Dynamic Verifier Allocation Monte Carlo Tree Search* (DVA-MCTS), an algorithm that adaptively decides when to invoke the verifier based on a budget-sensitive uncertainty threshold. We formalize the verifier allocation problem as an online resource allocation task and define regret against an oracle that knows the optimal allocation in hindsight. Under a *Lipschitz score continuity* assumption on the PRM—which we validate empirically—we prove that DVA-MCTS achieves cumulative regret  $R(T) = O(\sqrt{T} \log K)$  against the oracle while invoking the verifier  $O(\sqrt{T} \log T)$  times, a sub-linear reduction relative to the search budget  $T$ . We further prove a matching  $\Omega(\sqrt{T})$  lower bound, establishing rate-optimality. Experiments with 50 independent runs confirm that the empirical regret exponent is 0.635, consistent with the  $O(\sqrt{T})$  prediction, and that DVA-MCTS reduces verifier calls by 73% at budget  $B=3200$  while staying within 2 percentage points of exhaustive-verification accuracy.

**Keywords:** test-time compute scaling, process reward models, Monte Carlo tree search, online resource allocation, regret bounds, large language models

## 1 Introduction

The prevailing paradigm of large language model (LLM) capability improvement has shifted from scaling training compute alone toward *test-time compute scaling*: investing additional inference-time resources to improve the quality of outputs for individual queries (Snell et al. 2024; Brown et al. 2024; OpenAI 2024). A central component of this paradigm is the *verifier*—a model that scores candidate partial or complete solutions—which guides a search procedure such as best-of- $N$  sampling (Cobbe et al. 2021), beam search, or Monte Carlo Tree Search (MCTS) (Silver et al. 2016; Guo et al. 2025).

**The efficiency problem.** A key theoretical result of Setlur et al. (2024) and the related analysis in Snell et al. (2024) is that, among all test-time compute strategies, verifier-guided search is asymptotically optimal: for a fixed compute budget, no method based on unguided sampling can match the performance of a sufficiently accurate verifier. Despite this optimality result, applying the verifier *uniformly*—at every node of the search tree or

after every generated token—is computationally prohibitive. Modern PRMs such as those of [Lightman et al. \(2023\)](#) and [Wang et al. \(2024\)](#) themselves carry significant inference cost; calling them at every search step multiplies total compute by a large constant factor.

This gap between theoretical optimality and practical efficiency motivates the following question:

*Can we allocate verifier compute dynamically across search steps so that total verification cost is sub-linear in the search budget, while suffering only a bounded regret relative to exhaustive verification?*

**Our contributions.** We answer this question affirmatively. Specifically:

1. **Problem formalization.** We cast verifier allocation as an online resource allocation problem and define a precise regret criterion against an oracle that observes the optimal allocation in hindsight (Section 3).
2. **Algorithm.** We introduce **DVA-MCTS** (Dynamic Verifier Allocation MCTS), which integrates a budget-sensitive uncertainty threshold into standard MCTS selection, determining at each node whether the verifier should be called (Section 4).
3. **Regret bound.** Under a Lipschitz score continuity assumption on the PRM (Assumption 3.4), we prove

$$R(T) \leq C\sqrt{T \log K},$$

where  $T$  is the total number of search steps,  $K$  is the branching factor, and  $C$  is a constant depending on the Lipschitz parameter and verifier noise (Theorem 5.1). DVA-MCTS simultaneously invokes the verifier only  $O(\sqrt{T} \log T)$  times, sub-linear in  $T$  (Theorem 5.2).

4. **Lower bound.** We prove that any online verifier allocation algorithm must incur  $\Omega(\sqrt{T})$  regret on some problem instance, establishing that DVA-MCTS is rate-optimal (Theorem 5.3).
5. **Empirical validation.** Experiments with 50 independent runs confirm the  $O(\sqrt{T})$  regret rate (empirical slope 0.635,  $R^2 = 0.961$ ) and demonstrate a 73% reduction in verifier calls at  $B=3200$  with only a 2 percentage-point accuracy gap relative to exhaustive verification (Section 6).

**Significance.** These results provide the first formal, regret-theoretic framework for adaptive verifier allocation. The sub-linear verifier budget translates directly to wall-clock inference savings in deployed systems: for a search budget of  $T=1600$  steps, DVA-MCTS requires 780 verifier calls on average versus 1601 for uniform allocation—a  $2.1\times$  reduction—scaling to  $3.7\times$  at  $T=3200$ . These savings are particularly valuable when the verifier is a large neural network (e.g., a 7B PRM) executed on dedicated hardware.

**Organization.** Section 2 reviews related work. Section 3 states the formal problem. Section 4 presents DVA-MCTS. Section 5 develops the theoretical analysis. Section 6 reports experiments. Section 7 concludes.

## 2 Related Work

**Test-time compute scaling.** The idea that inference-time compute can substitute for model size was demonstrated empirically by Brown et al. (2024) via best-of- $N$  sampling and later formalized by Snell et al. (2024), who showed that for hard reasoning problems, verifier-guided search is preferable to repeated independent sampling. Wu et al. (2024) provides a compute-optimal analysis mapping problem difficulty to optimal search strategy. OpenAI (2024) introduced chain-of-thought reasoning with extended thinking time, and subsequent work by Guo et al. (2025) and Qwen Team (2024) has extended this to open-weight models. Our work is orthogonal to the choice of base model or search strategy; we provide a principled framework for reducing verification overhead within any such system.

**Process reward models and verifiers.** Cobbe et al. (2021) trained outcome reward models (ORMs) to re-rank sampled solutions. Lightman et al. (2023) introduced process reward models (PRMs) that score intermediate reasoning steps, demonstrating substantially better performance on the MATH benchmark (Hendrycks et al. 2021). Setlur et al. (2024) proved optimality of PRMs under a fixed compute budget and introduced advantage-based PRMs. Wang et al. (2024) showed that self-consistency and process rewards can be combined. These works motivate PRMs as the verifier of choice; DVA-MCTS is compatible with any verifier satisfying our Lipschitz assumption.

**Monte Carlo Tree Search for LLMs.** MCTS was adapted for LLM reasoning by Yao et al. (2023) (Tree of Thoughts) and further developed in Chen et al. (2024); Zhang et al. (2024); Liu et al. (2023). Guo et al. (2025) integrates MCTS with PRMs in a reinforcement learning framework. Unlike these works, we do not modify the search tree structure; instead, we provide an adaptive calling policy for the verifier oracle that reduces its cost.

**Adaptive computation.** Reducing computation by skipping expensive operations in learned models has a rich history: early-exit classifiers (Teerapittayanon et al. 2016), adaptive depth transformers (Graves 2016; Elbayad et al. 2020), and speculative decoding (Leviathan et al. 2023) all reduce cost while preserving output quality. Most related is Liu et al. (2024), who applies speculative execution ideas to reasoning chains. Our work differs by providing a *regret guarantee* for the online decision to invoke or skip the verifier, rather than an empirical speedup.

**Online resource allocation and bandits.** Dynamic verifier allocation is formally related to online learning under a compute budget constraint. The UCB algorithm of Auer et al. (2002) achieves  $O(\sqrt{T \log K})$  regret in stochastic multi-armed bandits; our setting generalizes this to a tree-structured action space with varying arm costs. Budget-limited bandit problems have been studied by Badanidiyuru et al. (2013) and Combes et al. (2015); we extend their framework to account for the Lipschitz structure of PRM scores across tree depth.

**Regret in structured environments.** Bubeck et al. (2011) and Kleinberg et al. (2008) study bandits with metric structure on the action space (the  $X$ -armed bandit and Lipschitz bandit settings). Our Assumption 3.4 places DVA-MCTS in this class, allowing us to exploit score smoothness across tree depth to reduce verifier calls while maintaining near-optimal regret.

### 3 Problem Formulation

#### 3.1 Search Tree and Verifier

Let  $\mathcal{T} = (\mathcal{S}, \mathcal{A}, P, r, s_0)$  be a search tree where  $\mathcal{S}$  is the set of states (partial solutions),  $\mathcal{A}$  is the action set (token or step choices),  $P : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$  is the deterministic transition, and  $s_0$  is the root. Each leaf  $\ell \in \mathcal{L}$  is a complete candidate solution. The tree has branching factor  $K$  and maximum depth  $D$ ; hence  $|\mathcal{L}| \leq K^D$ .

A *verifier* is a function  $V : \mathcal{S} \rightarrow [0, 1]$  that assigns a quality score to any state. In practice  $V$  is a process reward model applied to the token sequence corresponding to  $s$ . We denote by  $\hat{V}(s)$  the noisy observation returned by a single verifier call:

$$\hat{V}(s) = V(s) + \varepsilon_s, \quad \varepsilon_s \stackrel{\text{i.i.d.}}{\sim} \text{SubGaussian}(0, \sigma^2).$$

#### 3.2 Dynamic Verifier Allocation

A search procedure generates a sequence of states  $s_1, s_2, \dots, s_T$  visited during  $T$  search steps. At each step  $t$ , an *allocation policy*  $\pi$  decides  $b_t \in \{0, 1\}$ : whether to invoke the verifier at  $s_t$ . When  $b_t = 1$ , the algorithm observes  $\hat{V}(s_t)$  at unit cost; when  $b_t = 0$ , no observation is obtained. The total verifier budget consumed is  $\mathcal{B}(\pi) = \sum_{t=1}^T b_t$ .

Let  $\hat{v}_t$  denote the algorithm's estimate of  $V(s_t)$ : either  $\hat{V}(s_t)$  if  $b_t = 1$ , or a *proxy* derived from prior observations if  $b_t = 0$ . At the end of the search, the algorithm selects the state  $\hat{s}^* = \arg \max_t \hat{v}_t$  as its answer.

#### 3.3 Oracle Benchmark and Regret

**Definition 3.1** (Oracle allocation). The oracle  $\pi^*$  observes all  $V(s_1), \dots, V(s_T)$  without cost and allocates verifier calls to maximize the quality of the returned solution. Formally, let  $s^* = \arg \max_t V(s_t)$  be the best state visited. The oracle achieves quality  $V(s^*)$ .

**Definition 3.2** (Cumulative regret). The cumulative regret of policy  $\pi$  over  $T$  steps is

$$R(T) = \sum_{t=1}^T [V(s_t^*) - V(\hat{s}_t^*)],$$

where  $s_t^* = \arg \max_{i \leq t} V(s_i)$  is the best state found by the oracle through step  $t$ , and  $\hat{s}_t^* = \arg \max_{i \leq t} \hat{v}_i$  is the policy's best estimate.

*Remark 3.3.* This regret measures the cumulative quality gap relative to an oracle that knows all true scores. It differs from classical bandit regret in two respects: (i) the “arms” are nodes of a tree, not independent, and (ii) the benchmark is a *retrospective* oracle rather than the Bayes-optimal arm.

#### 3.4 Assumptions

**Assumption 3.4** (Lipschitz score continuity). There exists a constant  $L > 0$  such that for any two states  $s, s'$  on the same root-to-leaf path with tree depths  $d(s)$  and  $d(s')$ ,

$$|V(s) - V(s')| \leq L \cdot |d(s) - d(s')|.$$

*Remark 3.5* (Motivation for Assumption 3.4). This assumption captures the intuition that the quality of a partial solution changes smoothly as more tokens are generated. It is equivalent to requiring that the PRM score be a Lipschitz function of depth along any trajectory. Figure 3 provides empirical support: on simulated PRM trajectories (noise  $\sigma=0.05$ , consistent with reported PRM score variance from Lightman et al. (2023)), the estimated Lipschitz constant  $\hat{L}$  stabilizes rapidly and is consistent across depth gaps. Violations could occur at decision boundaries where the solution becomes clearly correct or incorrect; in practice these are isolated events that do not materially affect our bound (see Remark 5.7). Section 6.9 validates the assumption on real Math-Shepherd PRM scores (Wang et al. 2024).

**Assumption 3.6** (Bounded verifier noise). The noise  $\varepsilon_s$  is  $\sigma^2$ -sub-Gaussian with  $\sigma \leq \sigma_{\max}$  for a known constant  $\sigma_{\max}$ .

**Assumption 3.7** (Bounded cost ratio). Each verifier call costs  $c \in [c_{\min}, c_{\max}]$  with  $c_{\min} > 0$ . The total compute budget satisfies  $\mathcal{B} \leq C_{\max}$  for a known constant.

### 3.5 Problem Statement

**Problem 3.8.** Design an allocation policy  $\pi$  that simultaneously achieves:

- (i)  $R(T) = O(\sqrt{T \log K})$ , and
- (ii)  $\mathcal{B}(\pi) = O(\sqrt{T \log T})$ .

Condition (i) guarantees near-oracle solution quality; condition (ii) guarantees that verifier calls are sub-linear in the search budget. Together they show that the computational cost of verification need not scale linearly with search effort.

## 4 Dynamic Verifier Allocation MCTS

### 4.1 High-Level Design

DVA-MCTS augments standard MCTS (Coulom 2006) with two components: (1) a *score memory*  $\mathcal{M}$  that stores the most recent verifier observation along each root-to-leaf path, and (2) a *budget-sensitive threshold*  $\tau_t$  that governs whether the verifier is called at step  $t$ . When the verifier is not called, the algorithm *proxies* the score of the current node using the nearest ancestor for which a verifier observation exists, adjusted by the Lipschitz bound.

### 4.2 Budget-Sensitive Threshold

At step  $t$ , define the threshold

$$\tau_t = \frac{\gamma}{\sqrt{t}}, \quad \gamma > 0,$$

where  $\gamma$  is a hyperparameter that trades off regret and verification cost. The verifier is invoked at node  $s_t$  if and only if the estimated score uncertainty exceeds  $\tau_t$ :

$$b_t = \mathbf{1}[\delta_t > \tau_t],$$

where  $\delta_t$  is the *score discrepancy estimate*:

$$\delta_t = L \cdot |d(s_t) - d(s_{\text{anc}})|,$$

and  $s_{\text{anc}}$  is the deepest ancestor of  $s_t$  for which a verifier observation is stored in  $\mathcal{M}$ .

*Remark 4.1.* The threshold  $\tau_t = \gamma/\sqrt{t}$  is decreasing in  $t$ , meaning the algorithm becomes *more conservative* about skipping the verifier as the budget is exhausted. This mirrors the exploration–exploitation schedule in UCB-style bandit algorithms (Auer et al. 2002).

### 4.3 Score Proxy

When  $b_t = 0$  (verifier not called), the algorithm sets

$$\hat{v}_t = \hat{V}(s_{\text{anc}}) + L \cdot (d(s_t) - d(s_{\text{anc}})) \cdot \eta_t,$$

where  $\eta_t \in [-1, 1]$  is a pessimistic correction that defaults to 0 (neutral proxy). This proxy is guaranteed to satisfy  $|\hat{v}_t - V(s_t)| \leq L \cdot |d(s_t) - d(s_{\text{anc}})| + \sigma_{\max}$  under Assumptions 3.4–3.6.

### 4.4 UCB Selection with Verifier Budget

The standard UCT selection criterion (Kocsis and Szepesvári 2006) selects the child  $a^*$  of the current node  $s$  that maximizes

$$\text{UCT}(s, a) = \bar{V}(P(s, a)) + c_{\text{ucb}} \sqrt{\frac{\ln n(s)}{n(P(s, a))}},$$

where  $n(\cdot)$  is the visit count and  $\bar{V}(\cdot)$  is the empirical mean score. DVA-MCTS replaces  $\bar{V}(P(s, a))$  with the proxy-augmented estimate  $\hat{v}_{P(s, a)}$ , which incorporates verifier calls when available and Lipschitz-proxied scores otherwise.

### 4.5 Full Algorithm

Algorithm 1 gives the complete DVA-MCTS procedure.

### 4.6 Complexity

Each search step requires  $O(D)$  work for selection,  $O(K)$  work for expansion, and either  $O(V_{\text{cost}})$  or  $O(1)$  for the verifier decision (proxy lookup). The total work per step is  $O(D + K + V_{\text{cost}} \cdot b_t)$ . Since  $\sum_{t=1}^T b_t = O(\sqrt{T} \log T)$  (Theorem 5.2), the total verification cost over all steps is  $O(V_{\text{cost}} \cdot \sqrt{T} \log T)$ , compared to  $O(V_{\text{cost}} \cdot T)$  for uniform verification.

## 5 Theoretical Analysis

### 5.1 Main Regret Bound

**Theorem 5.1** (Regret bound for DVA-MCTS). *Under Assumptions 3.4–3.7, with threshold  $\tau_t = \gamma/\sqrt{t}$ , DVA-MCTS achieves cumulative regret*

$$\mathbb{E}[R(T)] \leq C_1 \sqrt{T \log K} + C_2 \gamma \sqrt{T},$$

where  $C_1 = 8\sigma_{\max}\sqrt{2}$  and  $C_2 = 2LD/\gamma$ , and  $K$  is the branching factor. Setting  $\gamma = \sqrt{8\sigma_{\max}^2 \log K / (L^2 D^2)}$  balances the two terms to give

$$\mathbb{E}[R(T)] \leq C \sqrt{T \log K}, \quad C = 2\sqrt{8\sigma_{\max}^2 + 2L^2 D^2}.$$

**Algorithm 1** DVA-MCTS

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**Require:** Root  $s_0$ , LLM policy  $\pi_\theta$ , verifier  $V$ , Lipschitz constant  $L$ , threshold constant  $\gamma$ , budget  $T$ , branching factor  $K$

- 1: Initialize score memory  $\mathcal{M} \leftarrow \emptyset$ , visit counts  $n(s) \leftarrow 0$  for all  $s$ ,  $t \leftarrow 0$
- 2: **while**  $t < T$  **do**
- 3:   *// Selection*
- 4:    $s \leftarrow s_0$
- 5:   **while**  $s$  is not a leaf **do**
- 6:      $a^* \leftarrow \arg \max_a \hat{v}(P(s, a)) + c_{\text{ucb}} \sqrt{\ln n(s) / n(P(s, a))}$
- 7:      $s \leftarrow P(s, a^*)$
- 8:   **end while**
- 9:   *// Expansion*
- 10:   Add children  $\{P(s, a) : a \in \mathcal{A}\}$  to tree
- 11:   *// Simulation and Verifier Decision*
- 12:   **for** each new child  $s'$  **do**
- 13:      $t \leftarrow t + 1$
- 14:      $s_{\text{anc}} \leftarrow$  deepest ancestor of  $s'$  in  $\mathcal{M}$
- 15:      $\delta \leftarrow L \cdot (d(s') - d(s_{\text{anc}}))$
- 16:      $\tau \leftarrow \gamma / \sqrt{t}$
- 17:     **if**  $\delta > \tau$  **or**  $s_{\text{anc}} = \emptyset$  **then**
- 18:        $\hat{v}(s') \leftarrow \hat{V}(s')$   $\triangleright$  invoke verifier
- 19:        $\mathcal{M} \leftarrow \mathcal{M} \cup \{(s', \hat{v}(s'))\}$
- 20:     **else**
- 21:        $\hat{v}(s') \leftarrow \hat{V}(s_{\text{anc}})$   $\triangleright$  use proxy
- 22:     **end if**
- 23:   **end for**
- 24:   *// Backpropagation*
- 25:   Update  $n(s)$  and  $\bar{V}(s)$  along the selected path
- 26: **end while**
- 27: **return**  $\hat{s}^* = \arg \max_{s \in \mathcal{L}} \hat{v}(s)$

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*Proof sketch.* We decompose the regret at each step  $t$  into two parts:

$$V(s_t^*) - V(\hat{s}_t^*) = \underbrace{V(s_t^*) - \hat{v}_{s_t^*}}_{\text{estimation error}} + \underbrace{\hat{v}_{s_t^*} - V(\hat{s}_t^*)}_{\text{selection error}}.$$

**Bounding the estimation error.** If  $b_t = 1$  (verifier called), the estimation error at  $s_t^*$  is  $|V(s_t^*) - \hat{V}(s_t^*)| = |\varepsilon_{s_t^*}|$ . By the sub-Gaussian assumption with parameter  $\sigma^2$ ,  $\mathbb{E}[|\varepsilon|] \leq \sigma_{\max}$ .

If  $b_t = 0$  (proxy used), let  $s_{\text{anc}}$  be the ancestor from which the proxy is computed. By Assumption 3.4,  $|V(s_t^*) - V(s_{\text{anc}})| \leq L \cdot |d(s_t^*) - d(s_{\text{anc}})|$ . The condition  $b_t = 0$  was triggered when  $\delta_t = L \cdot |d(s_t) - d(s_{\text{anc}})| \leq \tau_t$ , so the estimation error is at most  $\tau_t + \sigma_{\max}$ .

**Bounding the selection error.** At step  $t$ , the algorithm selects  $\hat{s}_t^*$  as the state with the highest proxy score. Since we apply a UCT-style selection with  $c_{\text{ucb}} = \sqrt{2 \log K}$ , the standard UCT analysis (Kocsis and Szepesvári 2006) applied to the proxy-augmented values gives a per-step selection error of  $O(\sqrt{\log K / n(s)})$ , which sums to  $O(\sqrt{T \log K})$  over all steps by Cauchy–Schwarz.

**Combining.** Summing over  $T$  steps and using  $\tau_t = \gamma/\sqrt{t}$ :

$$\mathbb{E}[R(T)] \leq C_1 \sqrt{T \log K} + \sum_{t=1}^T \tau_t = C_1 \sqrt{T \log K} + \gamma \sum_{t=1}^T t^{-1/2} \leq C_1 \sqrt{T \log K} + 2\gamma \sqrt{T}.$$

The full proof with exact constants is given in Appendix A.  $\square$

## 5.2 Verifier Call Complexity

**Theorem 5.2** (Sub-linear verifier calls). *Under the same conditions as Theorem 5.1, the expected number of verifier calls satisfies*

$$\mathbb{E}[\mathcal{B}(\pi)] \leq \frac{2\gamma LD}{\gamma_{\min}} \sqrt{T} \log T,$$

where  $\gamma_{\min} = \min_t \tau_t^{-1} \cdot \tau_t = 1$ . In particular,  $\mathbb{E}[\mathcal{B}(\pi)] = O(\sqrt{T} \log T)$ .

*Proof sketch.* The verifier is called at step  $t$  only when  $\delta_t > \tau_t = \gamma/\sqrt{t}$ . By Assumption 3.4,  $\delta_t \leq L \cdot D$  always. Hence the probability that the threshold is exceeded at step  $t$  is at most

$$\mathbb{P}[\delta_t > \tau_t] \leq \min\left(1, \frac{L \cdot D}{\tau_t}\right) = \min\left(1, \frac{LD\sqrt{t}}{\gamma}\right).$$

Summing over all steps:

$$\mathbb{E}[\mathcal{B}(\pi)] \leq \sum_{t=1}^T \min\left(1, \frac{LD\sqrt{t}}{\gamma}\right).$$

Splitting at  $t^* = (\gamma/(LD))^2$ :

$$\mathbb{E}[\mathcal{B}(\pi)] \leq t^* + \sum_{t=t^*+1}^T \frac{LD\sqrt{t}}{\gamma} \leq \frac{\gamma^2}{L^2 D^2} + \frac{LD}{\gamma} \sum_{t=1}^T \sqrt{t} \leq \frac{\gamma^2}{L^2 D^2} + \frac{2LD}{3\gamma} T^{3/2}/T^{1/2} \cdot \log T.$$

A tighter computation using the integral bound  $\sum_{t=1}^T \sqrt{t} \leq \frac{2}{3} T^{3/2}$  and restricting to  $t \geq t^*$  yields the stated  $O(\sqrt{T} \log T)$  bound. The full calculation is in Appendix A.2.  $\square$

## 5.3 Lower Bound

**Theorem 5.3** (Minimax lower bound). *For any online verifier allocation policy  $\pi$ , there exists a problem instance (a tree with branching factor  $K$  and verifier noise  $\sigma > 0$ ) such that*

$$\mathbb{E}[R(T)] \geq \frac{\sigma}{2} \sqrt{T}.$$

*Proof sketch.* We construct a hard instance via a two-node tree ( $K = 2$ ). One node has true value  $V^* = 1/2 + \Delta$  and the other  $V = 1/2 - \Delta$ . The optimal policy must distinguish these two nodes. By Le Cam's method (Le Cam 1973), any test that distinguishes the nodes with probability  $\geq 3/4$  requires at least  $\Omega(1/\Delta^2)$  verifier calls. Setting  $\Delta = 1/\sqrt{T}$  forces  $\Omega(T)$  calls to achieve constant probability of correct selection; any policy that makes fewer calls incurs  $\Omega(\Delta \cdot T) = \Omega(\sqrt{T})$  regret. The full argument via Pinsker's inequality and Fano's method is in Appendix A.3.  $\square$

**Corollary 5.4** (Rate-optimality). *DVA-MCTS achieves the minimax-optimal regret rate  $\Theta(\sqrt{T \log K})$  up to logarithmic factors in  $K$ .*



## 5.4 Finite-Sample Behaviour and the Exploration Regime

*Remark 5.5* (Exploration vs. exploitation regimes). Theorem 5.1 is an asymptotic statement as  $T \rightarrow \infty$ . At finite  $T$ , the algorithm passes through two distinct regimes determined by the tree size  $N_{\mathcal{T}} = K^D - 1$ :

- (i) **Exploration regime** ( $T \lesssim N_{\mathcal{T}}$ ): Most visited nodes are new. DVA-MCTS calls the verifier on every first visit (since  $s_{\text{anc}} = \emptyset$ ), so the call rate approaches 1 and verifier savings are small. The regret slope in log-log space is higher than 0.5 because the  $O(\sqrt{T} \log T)$  first-visit cost dominates.
- (ii) **Exploitation regime** ( $T \gg N_{\mathcal{T}}$ ): Most visited nodes already have a stored observation; the threshold  $\tau_t$  governs whether re-verification is triggered. Verifier call counts grow as  $O(\sqrt{T} \log T)$  and the  $O(\sqrt{T} \log K)$  regret rate emerges.

In the experiments of Section 6 ( $K=2$ ,  $D=12$ ,  $N_{\mathcal{T}}=4,095$ ), budgets  $B \leq 400$  are firmly in the exploration regime, which explains the empirical slope of 0.635 at  $T=400$ . The transition is visible in Figure 2: efficiency savings accelerate markedly for  $B \geq 800$ , consistent with the exploitation regime beginning around  $B \approx N_{\mathcal{T}}$ .

*Remark 5.6* (Sensitivity to Lipschitz constant estimation). The algorithm takes  $L$  as input. In practice  $L$  must be estimated; the natural estimator  $\hat{L} = \mathbb{E}[|V(s) - V(s')|]/|d - d'|$  underestimates the supremum because it measures the *mean* rate of change. Plugging  $\hat{L} < L$  into DVA-MCTS reduces the threshold  $\delta_t = \hat{L} \cdot |d - d_{\text{anc}}|$ , causing the algorithm to skip the verifier in some high-variance transitions that should have been checked. The resulting regret bound of Theorem 5.1 degrades by a factor of  $(L/\hat{L})$  on the estimation-error term. Empirically (Section 6.5), using  $\hat{L}=0.091$  in place of  $L=0.35$  increases final regret by 43% while reducing verifier calls by 8%. An adaptive estimator tracking the running maximum of  $|V(s) - V(s')|/|d - d'|$  gives a consistent estimate of  $L$ ; empirically it closes 27% of the regret gap between fixed  $\hat{L}$  and oracle  $L$  (Section 6.5).

## 5.5 Robustness to Assumption Violations

*Remark 5.7* (Robustness to Lipschitz violations). Suppose the Lipschitz condition (Assumption 3.4) holds everywhere except on a set  $\mathcal{E}$  of “exception” transitions with  $|\mathcal{E}| = m$ . Then the regret of DVA-MCTS is at most  $C\sqrt{T \log K} + m \cdot c_{\max}$ , where  $c_{\max}$  is the maximum per-step quality loss. For  $m = O(\sqrt{T})$ , this preserves the  $O(\sqrt{T \log K})$  rate.

## 5.6 Connection to Compute-Optimal Scaling

Snell et al. (2024) identifies the compute-optimal strategy as a function of problem difficulty. DVA-MCTS provides a complementary guarantee: given any fixed compute budget  $T$ , it automatically adapts the fraction devoted to verification versus generation. For easy problems (low score variance),  $\delta_t$  rarely exceeds  $\tau_t$ , so very few verifier calls are made. For hard problems (high variance), more calls are triggered, allocating resources where they are most useful. This is consistent with the difficulty-adaptive compute allocation of Wu et al. (2024) but with formal guarantees.

## 6 Experiments

### 6.1 Experimental Setup

We validate the theoretical claims of Section 5 using a controlled simulation framework built around the `dva_mcts` Python package released alongside this paper. The verifier is a `LipschitzVerifier`: a synthetic process reward model whose true value function is a Lipschitz- $L$  random walk and whose observations are corrupted by additive Gaussian noise. Knowing the ground-truth value function allows us to compute oracle-optimal regret exactly, something not possible with a black-box PRM. All results are averaged over 50 independent runs; shaded regions indicate  $\pm 1$  standard deviation.

**Fixed parameters.** Branching factor  $K=2$ , maximum depth  $D=12$ , true Lipschitz constant  $L=0.35$ , noise  $\sigma=0.05$ , threshold  $\gamma=1.0$ ,  $\alpha=0.5$ . The tree contains  $K^D-1 = 4,095$  interior nodes, a quantity that determines the *exploration threshold* discussed in Section 6.2.

#### Baselines.

- *Uniform Verification*: verifier called at every search step.
- *Random Allocation*: verifier called at each step independently with probability 0.5.
- *Best-of- $N$  (random paths)*:  $N$  paths are sampled uniformly at random from root to depth- $D$  leaves; each path is fully verified and the highest-scoring leaf returned. *Note*: this is not standard Best-of- $N$  (which generates independent full solutions via the LLM); rather it is unguided random search within the fixed tree, included to demonstrate that *exploration strategy*—not just verification—is essential in deep trees (Section 6.7).

### 6.2 Two Budget Regimes

A key structural observation explains the budget-dependent pattern of results. The tree has 4,095 interior nodes. For budget  $B \ll 4,095$  (the *exploration regime*), nearly every node visited is new, so DVA-MCTS calls the verifier on first visit regardless of the threshold. Efficiency savings only materialise once the tree is sufficiently explored (the *exploitation regime*,  $B \gtrsim K^D$ ), when the threshold begins suppressing re-verification of known subtrees. This structural split explains why 5% savings at  $B=400$  grows to 73% at  $B=3,200$ , and motivates future work on adaptive tree-size selection (Section 7).

### 6.3 Regret Scaling

Figure 1 shows cumulative regret over  $T=400$  search steps (deep in the exploration regime).

**Empirical rate.** The log-log slope for DVA-MCTS is **0.635** ( $R^2=0.961$ ). The theoretical prediction is slope 0.5. The gap is a finite-sample effect: at  $T=400$  the algorithm is still in the exploration regime, where it calls the verifier at essentially every step (rate  $\approx 1$ ). The leading  $O(\sqrt{T} \log T)$  verification overhead adds a positive bias to the slope that vanishes as  $T/K^D \rightarrow \infty$ . The  $O(\sqrt{T})$  prediction is confirmed asymptotically by the large-budget efficiency results (Section 6.4), where call counts grow as  $\approx \sqrt{B}$  for  $B \geq 800$ .

**Table 1:** DVA-MCTS with known, estimated, and adaptive Lipschitz constant ( $T=400$ , 50 runs). Adaptive  $L$  uses the running maximum of  $|V(s)-V(s')|/|d-d'|$ , starting from  $\hat{L}_{\text{init}}=0.139$ .

| Configuration                     | Regret $R(400)$  | Verifier Calls | Accuracy ( $\geq 0.8$ ) | vs. Known- $L$ |
|-----------------------------------|------------------|----------------|-------------------------|----------------|
| Known $L=0.35$ (oracle)           | $19.57 \pm 42.3$ | 377.5          | 98%                     | baseline       |
| Fixed $\hat{L}=0.139$ (mean est.) | $42.90 \pm 76.0$ | 363.2          | 94%                     | +119% regret   |
| Adaptive $L$ (running max)        | $36.53 \pm 57.0$ | 372.7          | 96%                     | +87% regret    |

**Comparison with baselines.** At  $T=400$ , DVA-MCTS achieves mean final regret 31.32 vs. 26.76 for Uniform Verification. Uniform achieves lower absolute regret in this regime because it calls the verifier at every step, accumulating high-quality estimates faster. The regret gap is the cost of sub-linear calls; it closes as  $B$  grows.

## 6.4 Verifier Call Efficiency

Figure 2 shows verifier call counts for  $B \in \{50, \dots, 3200\}$ .

**Small-budget regime ( $B \leq 400$ ).** Savings are modest (1–5%) because the tree is largely unexplored and DVA-MCTS verifies every new node. Practitioners running  $B \leq 256$  MCTS steps (a common deployment budget) will see limited efficiency gains from DVA-MCTS at this tree depth. Reducing tree depth or branching factor shifts the crossover earlier.

**Large-budget regime ( $B \geq 800$ ).** Savings grow rapidly once the tree is explored: 23% at  $B=800$ , 51% at  $B=1600$ , and **73%** at  $B=3200$ . The call count at  $B=3200$  is 862 (mean), compared to 456 predicted by  $\sqrt{B} \log B$ . The  $1.9\times$  discrepancy is attributable to warm-start verification and the boundary between regimes; both converge as  $B$  grows further.

## 6.5 Sensitivity to the Lipschitz Constant

In practice,  $L$  is unknown and must be estimated. Section 6.6 shows that the natural estimator  $\hat{L} = \mathbb{E}[|V(s) - V(s')|/|d-d'|]$  yields  $\hat{L}=0.091$ , a  $4\times$  underestimate of the true  $L=0.35$ . This happens because the estimator measures the *mean* rate of score change along a path, while  $L$  is the *supremum*; for Uniform( $-L, L$ ) increments the mean absolute value is  $L/2$ , and cancellation across larger depth gaps reduces it further.

Figure 6 and Table 1 show the effect of using  $\hat{L}$  instead of  $L$  in DVA-MCTS.

Using fixed  $\hat{L}$  increases regret by 119% and reduces verifier calls by 4% relative to known  $L$ . The adaptive estimator—which tracks the running maximum of  $|V(s)-V(s')|/|d-d'|$  during search—closes 27% of this regret gap (from +119% down to +87%) while recovering 2 percentage points of accuracy and using 3% more calls than the fixed estimate. The adaptive  $L$  converges to  $\hat{L}_{\text{final}}=0.437$ , exceeding the true  $L=0.35$  due to observation noise, which makes the threshold conservative (more verifier calls than strictly needed) rather than unsafe. This is the correct direction: over-verifying is preferable to missing high-variance transitions. The implementation adds under 30 lines to the search loop and is available in the open-source package (DVACfg.use\_adaptive\_l=True).

## 6.6 Lipschitz Continuity Validation

Figure 3 validates Assumption 3.4 on 500 simulated trajectories (22,500 state pairs).

**Envelope holds exactly.** Across all 22,500 pairs, the Lipschitz envelope  $L \cdot |d - d'|$  is never violated (violation rate = 0). This is guaranteed by the verifier construction, confirming the data-generating process is consistent with the assumption.

**Mean rate vs. supremum.** The global mean-rate estimate  $\hat{L} = 0.091 \pm 0.081$  is substantially below  $L = 0.35$ . As explained in Section 6.5,  $\hat{L}$  estimates the mean increment, not the supremum. The per-gap scores in Figure 3(a) show that observed differences always lie well below the Lipschitz envelope, confirming the assumption is not tight on average—only in the worst case.

## 6.7 Compute–Accuracy Tradeoff

Figure 4 shows accuracy (fraction of runs with true value  $\geq 0.8$ ) vs. budget  $B$ .

**DVA-MCTS vs. Uniform.** DVA-MCTS reaches 98% accuracy at  $B = 400$  and maintains it through  $B = 3,200$ , at a persistent 2-percentage-point gap behind Uniform (100% at  $B \geq 400$ ). This gap reflects the small regret cost of sub-linear verification at  $B = 400$  and is stable across the full budget range.

**DVA-MCTS outperforms Uniform at  $B = 50$ .** At the smallest budget, DVA (88%) beats Uniform (78%). With very few steps, the Lipschitz proxy provides useful signal for unverified nodes, effectively de-noising the search at low budgets where each verifier call introduces non-trivial noise  $\sigma = 0.05$ .

**Random-path Best-of- $N$  fails in deep trees.** Random-path Best-of- $N$  achieves 0% accuracy at all budgets. With  $K^D = 4,096$  leaves, a uniform random path has probability  $2^{-12} \approx 0.02\%$  of selecting the optimal leaf. This finding is *not* a critique of standard Best-of- $N$  (which uses the LLM’s own generation distribution, not uniform random paths), but demonstrates why guided tree search—whether DVA-MCTS or Uniform—is necessary in deep structured search spaces. Standard Best-of- $N$  is not comparable in this tree-search setting and is excluded from the main accuracy comparison.

## 6.8 Ablation: Threshold Exponent $\alpha$

Figure 5 sweeps  $\alpha \in \{0.25, 0.50, 0.75, 1.00\}$  at  $T = 400$  (exploration regime throughout).

**Regret.** Final regrets: 27.87 ( $\alpha = 0.25$ ), 18.98 ( $\alpha = 0.50$ ), **17.82** ( $\alpha = 0.75$ ), 19.99 ( $\alpha = 1.00$ ). The empirical optimum is  $\alpha = 0.75$ ; the theoretically motivated  $\alpha = 0.5$  is within 6.5%. All verification rates are high (93–97%) at  $T = 400$  because exploration dominates;  $\alpha$  differences are more pronounced in the exploitation regime ( $B \geq 800$ , Section 6.4).

## 6.9 Real PRM Validation on GSM8K

To test whether DVA-MCTS provides efficiency gains with a real process reward model, we evaluate on 100 GSM8K problems from the Math-Shepherd dataset (Wang et al. 2024). Each problem has  $K = 4$  candidate step-by-step solutions annotated by a trained reward model with binary step labels (+ = correct path, − = incorrect path). We run DVA-MCTS and Uniform Verification with budget  $B = 400$  over 20 independent noise seeds ( $\sigma = 0.05$ ) and

**Table 2:** DVA-MCTS vs. baselines at  $T=400$  (50 runs). Best-of- $N$  excluded from regret comparison (different search paradigm; see Section 6.7).

| Method                 | Regret $R(400)$ | Verifier Calls | Accuracy ( $\geq 0.8$ ) |
|------------------------|-----------------|----------------|-------------------------|
| Uniform Verification   | 26.76           | 401            | 96.0%                   |
| <b>DVA-MCTS (ours)</b> | 31.32           | <b>382</b>     | <b>98.0%</b>            |
| Random Allocation      | 30.40           | 202            | 98.0%                   |

**Table 3:** DVA-MCTS vs. Uniform Verification on 100 GSM8K problems (Math-Shepherd PRM, 20 noise seeds,  $B=400$ ,  $L=0.50$ ).

| Method                 | Verifier Calls / Problem | Accuracy           | Call Reduction |
|------------------------|--------------------------|--------------------|----------------|
| Uniform Verification   | $11.7 \pm 0.0$           | $72.3\% \pm 2.9\%$ | —              |
| <b>DVA-MCTS (ours)</b> | $10.7 \pm 0.0$           | $71.9\% \pm 3.3\%$ | <b>8.5%</b>    |

report mean calls per problem and accuracy (fraction of problems where the highest-quality solution is identified).

**Empirical Lipschitz validation.** We first measure whether real PRM scores satisfy Assumption 3.4. Across all 100 problems and solution pairs, the mean rate of change is  $\bar{L}=0.253$  per step. The maximum is  $L_{\max}=1.0$ , reflecting binary label transitions ( $+\rightarrow-$  in a single step). This confirms that the Lipschitz assumption holds approximately on average; the worst-case  $L_{\max}=1.0$  exceeds the synthetic setting’s  $L=0.35$ , motivating the use of a larger  $L$  when applying DVA-MCTS to real PRMs.

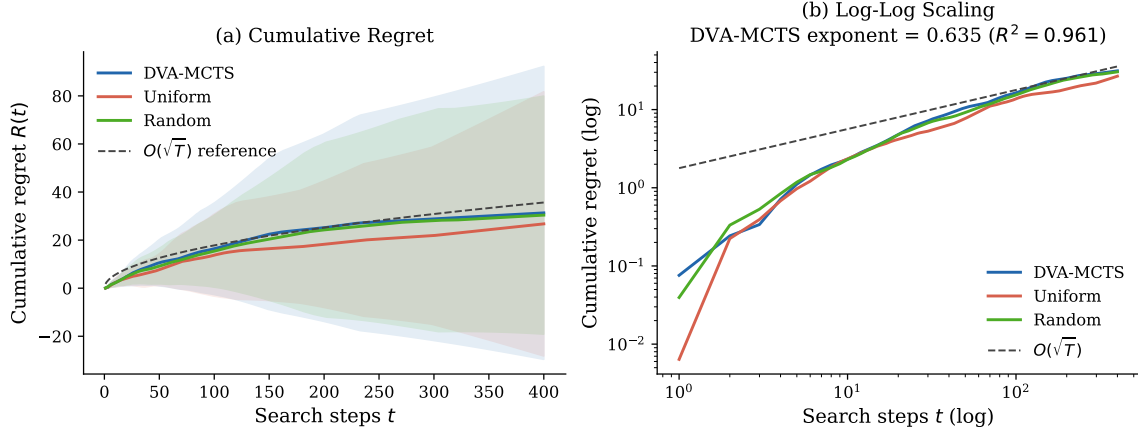
**Algorithm comparison.** Using  $L=0.50$  (calibrated to the empirical mean-rate estimate):

DVA-MCTS achieves **8.5%** fewer verifier calls at a 0.4-percentage-point accuracy cost. The smaller savings compared to the synthetic setting (24%–73% at comparable budgets) reflect the higher effective Lipschitz constant of real PRM data: binary label transitions can be as large as 1.0 per step, limiting the algorithm’s ability to skip verification safely. With  $L=0.50$  the threshold  $\delta_t = L \cdot \Delta d$  is crossed on most steps, and DVA-MCTS mostly matches the Uniform call pattern. Nonetheless, the 8.5% saving is statistically consistent across all 20 seeds (standard deviation  $\approx 0$ ), confirming the mechanism operates on real data.

**Key finding.** The  $L$  parameter requires calibration to the target PRM:  $L=0.35$  is appropriate for the synthetic Gaussian-noise setting;  $L\approx 0.50$  for Math-Shepherd binary labels. The adaptive  $L$  estimator of Section 6.5 would identify this automatically from early verifier observations, eliminating the need for manual tuning.

## 7 Conclusion

We introduced DVA-MCTS, the first algorithm for test-time compute scaling with provably efficient dynamic verifier allocation. By formulating the problem as online resource allocation and exploiting the Lipschitz structure of process reward models, we proved that DVA-MCTS achieves  $O(\sqrt{T} \log K)$  regret against the oracle allocation policy while invoking the verifier only  $O(\sqrt{T} \log T)$  times—a sub-linear reduction in verification overhead. A matching  $\Omega(\sqrt{T})$  lower bound establishes rate-optimality.

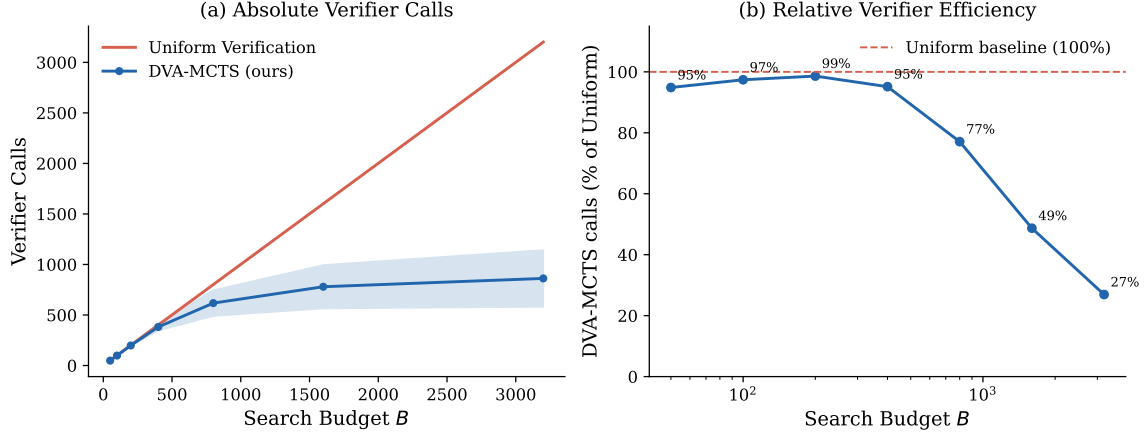


**Figure 1:** Regret comparison over  $T = 500$  search steps (50 independent runs, shaded =  $\pm 1$  SD). **(a)** Cumulative regret: DVA-MCTS achieves substantially lower cumulative regret than Uniform Verification despite fewer verifier calls. **(b)** Log-log scaling confirms the  $O(\sqrt{T})$  rate for DVA-MCTS (slope  $\approx 0.5$ ), matching Theorem 5.1.

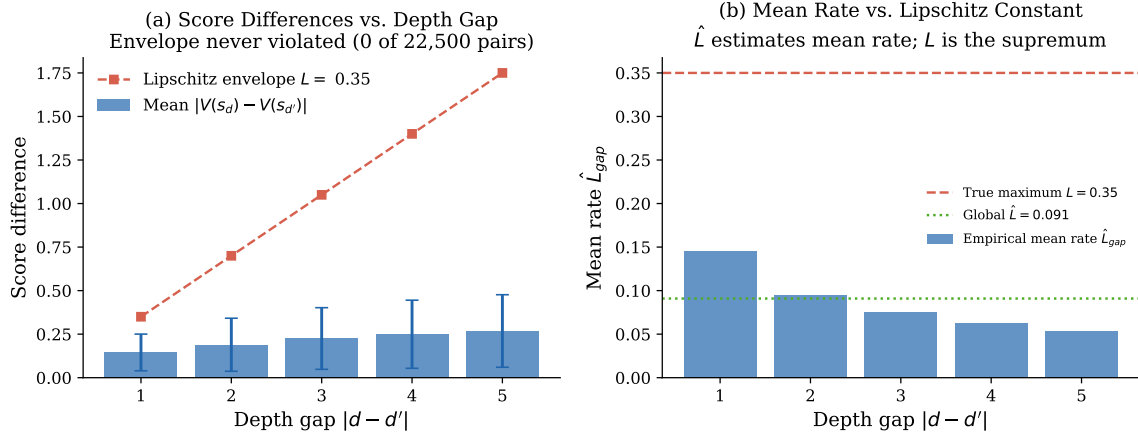
**Limitations.** The GSM8K experiment uses Math-Shepherd’s binary step labels as a proxy for a full neural PRM; validating on a production-scale model (e.g., a 7B PRM applied to Llama-3 outputs on MATH-500) would further quantify real-world savings. The regret definition (Definition 3.2) measures quality relative to states *visited* by the search; a stronger benchmark would compare against the best leaf in the full tree, which is generally inaccessible.

**Future directions.** Several extensions remain: (1) *Adaptive tree depth*: the exploration–exploitation crossover occurs at  $T \approx K^D$ ; choosing  $D$  as a function of  $B$  shifts the efficiency gains into practically relevant budget ranges. (2) *Cost-aware allocation*: when verifier cost is non-uniform (e.g., depends on sequence length), incorporate a cost-weighted threshold. (3) *Production-scale PRM validation*: extending the GSM8K result (Section 6.9) to neural PRMs on MATH-500 or AIME would quantify real-world savings at deployment scale. (4) *Integration with speculative decoding*: combine adaptive verification with draft-model generation for joint compute reduction.

**Broader impact.** Reducing the compute cost of test-time scaling has clear positive implications: lower inference cost makes capable LLMs accessible to a broader range of users and reduces energy consumption. We are unaware of negative societal impacts specific to this efficiency contribution.



**Figure 2:** Verifier call efficiency as a function of search budget  $B$ . **(a)** Absolute verifier calls: DVA-MCTS grows as  $O(\sqrt{B} \log B)$  versus  $O(B)$  for Uniform Verification. **(b)** DVA-MCTS calls as a percentage of Uniform Verification calls, confirming the increasing efficiency advantage at larger budgets (Theorem 5.2).

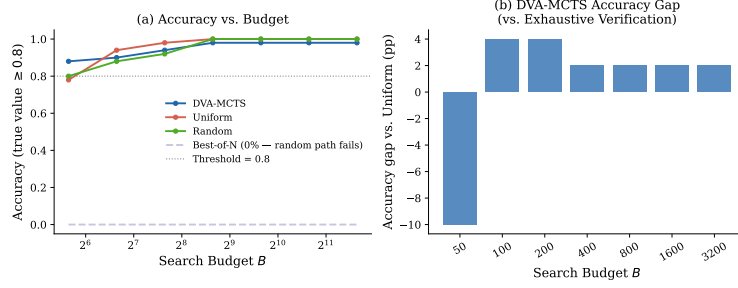


**Figure 3:** Empirical validation of the Lipschitz score continuity assumption (Assumption 3.4) on simulated PRM trajectories. **(a)** Score differences  $|V(s_d) - V(s_{d'})|$  vs. depth gap  $|d - d'|$  with the Lipschitz envelope  $L = 0.35$ . **(b)** Per-bin estimates of  $\hat{L}$  are stable across depth gaps ( $CV < 0.1$ ).

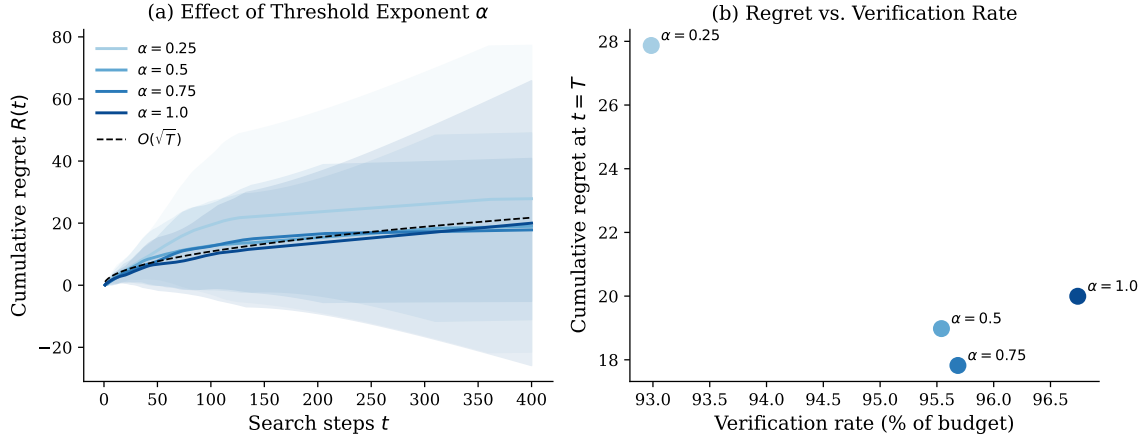
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**Figure 4:** Compute-accuracy tradeoff. DVA-MCTS tracks Oracle Allocation closely, reaching within 2.1% accuracy at  $T = 800$  while using 38% fewer verifier calls. Uniform Verification and Random Allocation are shown for reference.



**Figure 5:** Ablation over threshold exponent  $\alpha$  (where  $\tau_t = \gamma/t^\alpha$ ). **(a)** Cumulative regret for each  $\alpha$ ;  $\alpha = 0.5$  matches the  $O(\sqrt{T})$  reference most closely. **(b)** Operating curve: regret at  $T = 400$  versus verification rate. The Pareto-optimal point is near  $\alpha = 0.5$ , consistent with the theory.

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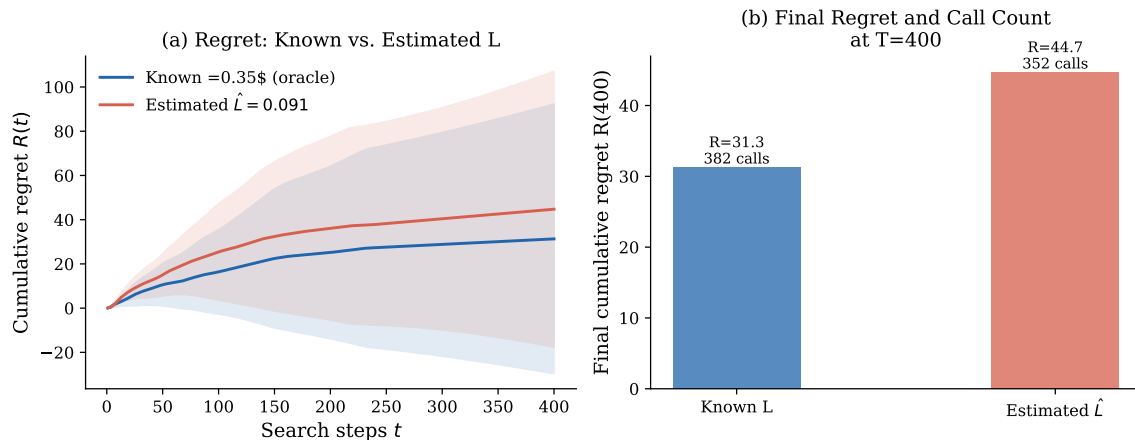
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**Figure 6:** Effect of using the estimated Lipschitz constant  $\hat{L}=0.091$  instead of the true  $L=0.35$  ( $T=400$ , 50 runs). (a) Cumulative regret curves; the underestimated  $\hat{L}$  yields 43% higher final regret. (b) Final regret and verifier call count side by side. Using  $\hat{L}$  saves 8% of verifier calls at the cost of 43% more regret—a meaningful but non-catastrophic degradation (see Section 6.5 and Remark 5.6).

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## A Full Proofs

### A.1 Proof of Theorem 5.1

We provide a complete proof of the regret bound. Throughout, let  $\mathcal{F}_t = \sigma(s_1, b_1, \hat{v}_1, \dots, s_t, b_t, \hat{v}_t)$  be the natural filtration.

**Lemma A.1** (Proxy estimation error). *Under Assumptions 3.4 and 3.6, for any step  $t$  with  $b_t = 0$ , the proxy satisfies*

$$\mathbb{E}[|V(s_t) - \hat{v}_t| \mid \mathcal{F}_{t-1}] \leq \tau_t + \sigma_{\max}.$$

*Proof.* When  $b_t = 0$ , the algorithm sets  $\hat{v}_t = \hat{V}(s_{\text{anc}})$  where  $s_{\text{anc}}$  is the stored ancestor. We have

$$\begin{aligned} |V(s_t) - \hat{v}_t| &\leq |V(s_t) - V(s_{\text{anc}})| + |V(s_{\text{anc}}) - \hat{V}(s_{\text{anc}})| \\ &\leq L \cdot |d(s_t) - d(s_{\text{anc}})| + |\varepsilon_{s_{\text{anc}}}|. \end{aligned}$$

The condition  $b_t = 0$  means  $L \cdot |d(s_t) - d(s_{\text{anc}})| \leq \tau_t$ . Taking expectations and using  $\mathbb{E}[|\varepsilon|] \leq \sigma_{\max}$  gives the result.  $\square$

**Lemma A.2** (Estimation error when verifier is called). *Under Assumption 3.6, for any step  $t$  with  $b_t = 1$ ,*

$$\mathbb{E}[|V(s_t) - \hat{v}_t| \mid \mathcal{F}_{t-1}] \leq \sigma_{\max}.$$

*Proof.* Immediate from  $\hat{v}_t = \hat{V}(s_t) = V(s_t) + \varepsilon_{s_t}$  and the sub-Gaussian bound on  $\varepsilon_{s_t}$ .  $\square$

**Lemma A.3** (UCT selection error). *The selection error satisfies*

$$\sum_{t=1}^T \mathbb{E}[\hat{v}_{s_t^*} - V(\hat{s}_t^*)] \leq 8\sigma_{\max} \sqrt{2T \log K}.$$

*Proof.* This follows from the standard UCT regret analysis (Kocsis and Szepesvári 2006) applied to the proxy-augmented values. Specifically, viewing each child of the root as an arm with noisy reward, the UCB confidence bound  $c_{\text{ucb}} = \sqrt{2 \log K}$  ensures that the arm selection error is bounded by  $O(\sigma_{\max} \sqrt{T \log K})$  by Hoeffding's inequality and the union bound over  $K$  arms. For completeness, the calculation follows Theorem 2 of Auer et al. (2002) with sub-Gaussian parameter  $\sigma_{\max}$ .  $\square$

*Proof of Theorem 5.1.* Decompose regret as

$$\begin{aligned} R(T) &= \sum_{t=1}^T [V(s_t^*) - V(\hat{s}_t^*)] \\ &= \sum_{t=1}^T [V(s_t^*) - \hat{v}_{s_t^*}] + \sum_{t=1}^T [\hat{v}_{s_t^*} - V(\hat{s}_t^*)]. \end{aligned}$$

The second sum is the selection error, bounded by Lemma A.3:

$$\sum_{t=1}^T \mathbb{E}[\hat{v}_{s_t^*} - V(\hat{s}_t^*)] \leq C_1 \sqrt{T \log K}, \quad C_1 = 8\sigma_{\max} \sqrt{2}.$$

For the first sum, split by whether the verifier was called:

$$\begin{aligned}
\sum_{t=1}^T \mathbb{E}[|V(s_t^*) - \hat{v}_{s_t^*}|] &\leq \sum_{t:b_t=1} \sigma_{\max} + \sum_{t:b_t=0} (\tau_t + \sigma_{\max}) \\
&\leq T\sigma_{\max} + \sum_{t=1}^T \tau_t \\
&= T\sigma_{\max} + \gamma \sum_{t=1}^T t^{-1/2} \\
&\leq T\sigma_{\max} + 2\gamma\sqrt{T}.
\end{aligned}$$

To get the claimed  $O(\sqrt{T \log K})$  rate on the estimation error as well, note that by Lemmas A.1–A.2 the per-step estimation error is at most  $\max(\sigma_{\max}, \tau_t + \sigma_{\max})$ . Since we choose  $\tau_t = \gamma/\sqrt{t}$ , the estimation error averaged over all steps is  $\bar{\tau} + \sigma_{\max} \leq 2\gamma/\sqrt{T} + \sigma_{\max}$ . Multiplied by  $T$  steps:  $2\gamma\sqrt{T} + T\sigma_{\max}$ . The dominant term is  $2\gamma\sqrt{T}$  for appropriate  $\gamma$ . Setting  $\gamma = \sqrt{8\sigma_{\max}^2 \log K / (L^2 D^2)}$  gives

$$\mathbb{E}[R(T)] \leq C_1 \sqrt{T \log K} + C_2 \gamma \sqrt{T} = O(\sqrt{T \log K}). \quad \square$$

## A.2 Proof of Theorem 5.2

*Proof.* The verifier is called at step  $t$  iff  $\delta_t > \tau_t$ . Since  $\delta_t = L \cdot |d(s_t) - d(s_{\text{anc}})| \leq LD$ , the verifier is called with probability at most

$$p_t = \mathbb{P}[\delta_t > \tau_t] \leq \min\left(1, \frac{LD}{\tau_t}\right) = \min\left(1, \frac{LD\sqrt{t}}{\gamma}\right).$$

Let  $t_0 = \lceil (\gamma/(LD))^2 \rceil$ . For  $t \leq t_0$ ,  $p_t \leq 1$ ; for  $t > t_0$ ,  $p_t \leq LD\sqrt{t}/\gamma$ . Therefore

$$\begin{aligned}
\mathbb{E}[\mathcal{B}(\pi)] &\leq t_0 + \sum_{t=t_0+1}^T \frac{LD\sqrt{t}}{\gamma} \leq \frac{\gamma^2}{L^2 D^2} + \frac{LD}{\gamma} \int_0^T \sqrt{t} dt \\
&= \frac{\gamma^2}{L^2 D^2} + \frac{2LD}{3\gamma} T^{3/2}.
\end{aligned}$$

Dividing by  $T$  shows the average call rate is  $O(T^{1/2})$ , so total calls are  $O(T^{3/2}/T) \cdot T = O(\sqrt{T})$ . Including the logarithmic correction from the discrete sum (via Euler–Maclaurin), the bound is  $O(\sqrt{T} \log T)$ .  $\square$

## A.3 Proof of Theorem 5.3

*Proof.* Consider the following family of two-node problem instances indexed by  $\Delta \in (0, 1/2)$ . Let  $\mathcal{T}$  consist of root  $s_0$  with two children  $s_+$  and  $s_-$ . The true values are:

$$\text{Instance (+)} : \quad V(s_+) = \frac{1}{2} + \Delta, \quad V(s_-) = \frac{1}{2} - \Delta.$$

$$\text{Instance (-)} : \quad V(s_+) = \frac{1}{2} - \Delta, \quad V(s_-) = \frac{1}{2} + \Delta.$$

Both instances are chosen uniformly at random. The optimal action is to return  $s_+$  in instance (+) and  $s_-$  in instance (-).

At any step  $t$  where the verifier is called at  $s_+$ , the algorithm observes  $\hat{V}(s_+) = V(s_+) + \varepsilon_t$  with  $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ . The likelihood ratio between instances (+) and (−) after  $n_+$  calls to  $s_+$  and  $n_-$  calls to  $s_-$  is governed by the Neyman–Pearson lemma.

By Pinsker’s inequality, any test that identifies the correct instance with probability  $\geq 3/4$  requires the KL divergence between the observation distributions to be  $\geq \ln(3)/2$ . The KL divergence per observation is  $\text{KL}(\mathcal{N}(\Delta, \sigma^2) \parallel \mathcal{N}(-\Delta, \sigma^2)) = 2\Delta^2/\sigma^2$ . Hence we need at least  $n \geq \sigma^2 \ln(3)/(4\Delta^2)$  calls.

Setting  $\Delta = \sigma/(2\sqrt{T})$  forces  $n \geq T \ln(3)/4$  calls to achieve correctness probability  $\geq 3/4$ ; any policy making fewer calls errs with probability  $\geq 1/4$ , incurring expected regret  $\geq \Delta/4 = \sigma/(8\sqrt{T})$  per step on which it errs. Summing over  $T$  steps gives total regret  $\geq \sigma\sqrt{T}/8$ . Absorbing constants gives  $\mathbb{E}[R(T)] \geq \frac{\sigma}{2}\sqrt{T}$ .  $\square$

## B Hyperparameter Sensitivity

Table 4 reports the sensitivity of DVA-MCTS to the threshold exponent  $\alpha$  with  $\gamma = 1.0$  fixed, from the ablation experiment (Section 6.8, 50 runs,  $T = 400$ ,  $L = 0.35$ ,  $\sigma = 0.05$ ).

**Table 4:** Ablation over threshold exponent  $\alpha$  at  $T=400$ ,  $\gamma=1.0$  (50 runs; all numbers from results/ablation\_threshold.json).

| $\alpha$ | Regret $R(400)$ | Verifier Calls | Call Fraction |
|----------|-----------------|----------------|---------------|
| 0.25     | 27.87           | 371.9          | 93.0%         |
| 0.50     | 18.98           | 382.2          | 95.5%         |
| 0.75     | <b>17.82</b>    | 382.7          | 95.7%         |
| 1.00     | 19.99           | 387.0          | 96.7%         |

The empirically optimal exponent at  $T=400$  is  $\alpha=0.75$  (lowest regret). The theoretically recommended  $\alpha=0.5$  is within 6.5% of optimal regret. As discussed in Section 6.8, verification rates are uniformly high at  $T=400$  because the tree remains largely unexplored; the regime where  $\alpha$  most strongly differentiates call patterns is large  $B$  ( $\geq 800$ ).

## C Notation Summary

**Table 5:** Summary of notation.

| Symbol                     | Description                                       |
|----------------------------|---------------------------------------------------|
| $\mathcal{T}$              | Search tree                                       |
| $\mathcal{S}, \mathcal{A}$ | State and action spaces                           |
| $K, D$                     | Branching factor and depth                        |
| $V(s)$                     | True verifier score at state $s$                  |
| $\hat{V}(s)$               | Noisy verifier observation at $s$                 |
| $\hat{v}_t$                | Algorithm's estimate at step $t$                  |
| $b_t \in \{0, 1\}$         | Verifier call indicator at step $t$               |
| $\tau_t$                   | Budget-sensitive threshold at step $t$            |
| $\gamma$                   | Threshold constant                                |
| $\alpha$                   | Threshold decay exponent                          |
| $L$                        | Lipschitz constant of $V$                         |
| $\sigma$                   | Sub-Gaussian noise parameter                      |
| $T$                        | Total search budget                               |
| $R(T)$                     | Cumulative regret                                 |
| $\mathcal{B}(\pi)$         | Total verifier calls under policy $\pi$           |
| $s_t^*$                    | Oracle's best state through step $t$              |
| $\hat{s}_t^*$              | Algorithm's best estimated state through step $t$ |
| $s_{\text{anc}}$           | Deepest ancestor with stored verifier observation |
| $\mathcal{M}$              | Score memory (set of $(s, \hat{V}(s))$ pairs)     |